

Isaac Nathan da Silva Barbosa

Full-Stack AI Engineer · Computer Engineering · AI · Cloud · IoT

Build intelligent systems that connect AI, cloud, and product — from agentic pipelines to production deploy.

PWD — ASD (ICD-11: 6A02.2) + ADHD (ICD-11: 6A05.2)

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TECHNICAL SKILLS

AI / ML: GPT-4.1 · Azure OpenAI · PyTorch · DeepForest · OpenCV · Ollama · DALL-E 3 · Flux · Sora · Veo 3.1 · Gemini SDK · scikit-learn

Backend: Python · FastAPI · Node.js · Express · Java · Azure Cosmos DB · PocketBase · SQL

Frontend: React 19 · TypeScript · JavaScript · Flutter · Tailwind CSS v4 · Vite · React Native (Expo 54)

Cloud & Infra: Azure · Docker · Git / GitHub · PocketBase · Linux (CachyOS/Hyprland) · PyInstaller / Inno Setup

Integrations: Mercado Pago PIX · Canva Connect API · Placid API · Templated.io · Pexels API · NVIDIA NIM API

Other: Fluent English · Cybersecurity · Agile Methodologies · API Testing · NiceGUI · Rich CLI · xlwings/COM

PROFESSIONAL SUMMARY

From Industrial Chemistry to Computer Engineering: first Python in 2022 (Thonny IDE, no GPT, extra afternoon classes). Built ForestAI from scratch, manually annotating hundreds of drone images from Fundação Renova. At Paware, migrated hundreds of GB to Azure Cosmos DB (Meritage Homes, USA) and architected agentic AI pipelines for HelloSocial (GPT-4.1 tool calling, DALL-E 3, Flux). I learn by solving real problems — from cross-platform MIME types to ReAct agents with schema enforcement.

PROFESSIONAL EXPERIENCE

Full-Stack Developer with AI focus | Oct 2025 — May 2026

Paware Softwares | Remote

- Migrated hundreds of GB of legacy datasets from Google Drive to Azure Cosmos DB (Meritage Homes, USA) — automated extraction, compression, injection into panels for WhatsApp agent. Docker + schema validation with auto-rollback.
- Resolved cross-platform MIME type issue (Android vs iOS) — validation layer that became the backbone of the Cosmos DB migration, with extensive testing on iOS devices.
- Architected agentic image-generation pipeline for HelloSocial — GPT-4.1 agent with tool calling (up to 10 iterations), Flux Kontext Pro + DALL-E 3 + Placid/Canva, local SQLite/PIL fallback.

Robotics Instructor | Sep 2025 — Oct 2025

SuperNerds | Mariana, MG

- Learned Arduino and LEGO in under 2 weeks and taught robotics to children/teens.
- Balanced mornings here with evening work at Paware — discipline and adaptability.

EDUCATION

Computer Engineering | 2024 — 2029

Cruzeiro do Sul

5th semester — in progress

Chemical Engineering | 2022 — 2024

UFSJ — Campus Alto Paraopeba

Transition — self-directed AI study

Industrial Chemistry | 2022

UFOP

Where Python began — Thonny IDE

PROJECTS

HarplA · github.com/xAngryBadger/harpia

Python · GPT-4.1 · DALL-E 3 · Flux 2.0 Pro · Sora · Veo 3.1 · Gemini SDK · Azure Cosmos DB · SQLite · PIL

- Creative automation engine with 7+ AI models — autonomous agentic pipeline: GPT-4.1 orchestrates tool calls (image generation, Pexels search, copywriting, compositing) with schema enforcement, up to 10 reasoning iterations (think -> call -> observe).
- Lightweight stack by default: SQLite + local PIL with 8 custom templates; falls back to paid APIs (DALL-E 3, Flux, Sora, Veo 3.1) when needed.
- Backend swap via env var: local SQLite ↔ Azure Cosmos DB + Blob Storage. Cron-ready with file-locking and stuck batch recovery. 6,930+ LOC async Python, zero hardcoded secrets.

Flora Sensus · github.com/xAngryBadger/flora-sensus

Flutter · Dart · Drift/SQLite · React · PocketBase · TypeScript

- Flutter offline-first forest inventory app — custom sync engine: cascading UUID remapping through FK chain (Propriedade -> UT -> Parcela -> Planta -> Foto), atomic rollback via Drift. ~24K LOC, sync architecture built from scratch.
- React admin panel with auth, photos, reports, XLSX/PDF/CSV export. Exponential backoff with jitter, auth retry with transparent 401 refresh.

ForestAI · github.com/xAngryBadger/forestai

PyTorch · DeepForest · OpenCV · scikit-learn · TensorBoard

- Forest species detection with Deep Learning — built from scratch without AI-assisted coding. Manual annotation of hundreds of drone images (Fundação Renova), DeepForest/YOLO training on local GPU, stratified splits, TensorBoard for memorization vs generalization analysis.

Fennec Excel · github.com/xAngryBadger/Sahara-Fenneck

Python · Ollama/qwen2.5 · CustomTkinter · xlwings/COM · PyInstaller · Inno Setup

- Local AI assistant for Excel via ReAct agent (Ollama/qwen2.5) — natural language to filter/sort/manipulate spreadsheets with auto-checkpoint. 6+ OAuth integrations (Gmail, Teams, Calendar, Drive, Outlook, Trello). Native Windows installer.

SRF System · github.com/xAngryBadger/srf-system

Python · pandas · openpyxl · NiceGUI · Rich CLI · unittest

- Planning engine for forest restoration — automatic executive dossiers with crew allocation, territories, and schedules. NiceGUI + Rich CLI.

Inovesa Florestal

React 19 · Motion · Lenis · Tailwind CSS v4 · TypeScript · Vite 6

- Premium website for forestry company — parallax, smooth scroll, multi-step ombudsman, cinematic transitions. Reusable motion system (7 variants, 3 springs). prefers-reduced-motion.

MaineCoon · github.com/xAngryBadger/minneapolis

Node.js · mineflayer · NVIDIA NIM API · Reinforcement Learning · pathfinder

- Minecraft bot with natural language commands via LLM — mine, craft, navigate. Reinforcement learning module for autonomous behavior.

HelloSocial

Python · FastAPI · Azure OpenAI · Flux · Canva Connect · React · TypeScript

- AI-powered social media platform — project at Paware that inspired HarplA. Image generation pipeline with Flux Kontext Pro and DALL-E 3.

CERTIFICATIONS

- **Python Essentials 1 & 2** — Cisco Networking Academy — Fundamentals + advanced Python (OOP, libraries, cert prep).
- **Data Science Essentials with Python** — Cisco Networking Academy — Pandas, Matplotlib — hands-on project-based learning.
- **Data Analytics Essentials** — Cisco Networking Academy — Industry-recognized analytics tools.
- **AI for Process Optimization** — Escola Virtual Gov · Enap · Serpro (71h) — Gov AI Nucleus (PBIA) — strategic AI in public management and infosec.
- **Networking Basics (120h)** — Cisco Networking Academy — Completed during Computer Engineering.
- **Intro to Cybersecurity** — Cisco Networking Academy — Network security and cyber threats.
- **IT Security** — Bradesco Foundation — Corporate infrastructure and data protection.
- **Fluent English (3 years)** — KUMON — Technical documentation and meetings with international teams.

LANGUAGES

Portuguese — Native · English — Fluent (KUMON, 3 years)

Full case studies, demos & source code -> xangrybadger.github.io/isaac-vitae